

AI Foundations | AIFo

Summary

CONTENTS

1. Machine Learning (ML)

2

2. Probability

2

2.1. Random Variables

2

2.2. Discrete random variables

2

2.2.1. Multiple Random Variables

3

2.3. Conditional Probability

3

2.4. Continuous random variables

4

2.5. Distributions

6

2.5.1. Univariate normal distribution

6

2.5.2. Bernoulli distribution

7

3. Tree Diagrams and sequential events

8

1. MACHINE LEARNING (ML)

There are many different AI research fields and AI applications. Algorithms and applications where a computer learns from data are typically in the field of statistical machine learning (ML).

Computer: performs a mapping from input to output.

Processing: any sort of non-trivial “calculation” (mapping) or information processing

Input: any sort of data/information (sensory input, bits, mechanical configuration...)

Output: any sort of response (data, actions, new state of a system...)

Data Visualization: Gives intuitive understanding of structure in the data, helps in identifying patterns, detecting outliers and data quality. Raw data → Information

TODO:

plot examples

<https://ourworldindata.org/>

<https://informationisbeautiful.net/>

2. PROBABILITY

2.1. RANDOM VARIABLES

A random variable X is a function from the sample space to the real numbers.

$$X : S \rightarrow \mathbb{R}$$

Random Variables come in two flavours:

discrete: X takes any of a finite or countable set of values, e.g. $\{-8, 1.5, 1.5, 2.693, 5, 6.3, 10\}$

continuous: X takes any value of an uncountable range, e.g. the real numbers in the interval $(2, 7)$.

$\Pr(X = x)$ is the probability that the random variable X takes the value x . It's often written as $P(x)$ or $p(x)$ or $P_X(x)$ or $\mathbb{P}(x)$.

TODO:

MathFML explained this a lot better

Definition 1: Range

The range of a random variable X , shown by $\text{Range}(X)$ or R_X , is the set of possible values of X .

2.2. DISCRETE RANDOM VARIABLES

The Probability Mass Function (PMF) of a discrete random variable is a function $P(x)$ that provides the probability for each value x of a discrete random variable X .

TODO:

$$0 \leq P_X(x) \leq 1$$

$$\sum_{x \in R_X} P_X(x) = 1$$

$$\text{for any set } A \subset R_X, P(X \in A) = \sum_{x \in A} P_X(x)$$

Example 1:

Dice rolls

Value x of the random Variable X	1	2	3	4	5	6
$P(X = x)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

From the PMF, we can calculate the expected value

$$E[X] = \sum_{i=1}^{\infty} P(x_i) \cdot x_i$$

2.2.1. Multiple Random Variables

The (joint) properties of multiple random variables are defined by the **Joint Probability Mass Function**, usually simply called Joint Probability.

For **independent** random variables, the joint probability is simply the product of the individual probabilities

$$P(X \cap Y) = P(X) \cdot P(Y)$$

Example 2: Rolling two dice

The random experiment of rolling die 1 and die 2 together has 36 possible outcomes (events). The Joint Probability is:

$$P(X = 5, Y = 4) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

	$X = 1$	$X = 2$	$X = 3$	$X = 4$	$X = 5$	$X = 6$
$Y = 1$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
$Y = 2$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
$Y = 3$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
$Y = 4$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
$Y = 5$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
$Y = 6$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$

2.3. CONDITIONAL PROBABILITY

For **dependent** random variables we use the conditional probability of X given Y defined as

$$P(X|Y) = \frac{P(X \cap Y)}{P(Y)}, \text{ when } P(Y) > 0$$

TODO:

$$P(X \cap Y) = P(X|Y) \cdot P(Y)$$

$$X \cap Y = \emptyset \Rightarrow P(X|Y) = 0$$

$$Y \subset X \Rightarrow P(X|Y) = 1$$

$$X \subset Y \Rightarrow P(X|Y) = \frac{P(X)}{P(Y)}$$

$$P(X \cap Y) = P(X)P(Y|X) = P(Y)P(X|Y)$$

Independence: Two events X and Y are independent iff $P(X \cap Y) = P(X)P(Y)$

Law of Total Probability: If $Y_1, Y_2, Y_3, \dots$ is a partition of the sample space S , then for any event X we have

$$P(X) = \sum_i P(X \cap Y_i) = \sum_i P(X|Y_i)P(Y_i)$$

Bayes' Rule: For any two Events X and Y , where $P(X) \neq 0$, we have

$$P(Y|X) = \frac{P(X|Y) \cdot P(Y)}{P(X)}$$

$$\text{Interpretation: } \text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Normalizer}}$$

Conditional independence: Two events X and Y are conditionally independent given an event Z with $P(Z) > 0$ if

$$P(X \cap Y|Z) = P(X|Z)P(Y|Z) \\ \Rightarrow P(X|Y \cap Z) = P(X|Z)$$

Example 3: Chance of rain

Let X be the event to observe clouds (0=no clouds, 1=small clouds, 2=big clouds) and Y the event that it rains (0=no rain, 1=light rain, 2=moderate rain, 3=heavy rain)

TODO:

Given a known joint distribution of two discrete random variables, say, X and Y , the marginal distribution of either variable – X for example – is the probability distribution of X when the values of Y are not taken into consideration. This can be calculated by summing the joint probability distribution over all values of Y .

	$X = 0$	$X = 1$	$X = 2$
$Y = 0$	0.35	0.21	0.03
$Y = 1$	0.10	0.07	0.04
$Y = 2$	0.00	0.05	0.05
$Y = 3$	0.00	0.02	0.08

$$P(X) = \sum_Y P(X, Y)$$

$$P(Y) = \sum_X P(X, Y)$$

2.4. CONTINUOUS RANDOM VARIABLES

For a continuous random variable X following holds:

$$P(-\infty < x < \infty) = 1$$

$$P(a < x < b) = \int_a^b f(x) dx$$

$$P(x) = \emptyset$$

The **uniform probability distribution** $\text{unif}(\emptyset, 1)$ characterizes an experiment in which one number is chosen from the sample space $\Omega = [0; 1]$ in such a way, that all numbers of Ω have an equal chance to occur.

Whenever $\Omega \subset \mathbb{R}$, we can use $\mathbb{P}$ to define the **cumulative distribution function** (CDF)

$$\begin{aligned} F(\alpha) &= \mathbb{P}((-\infty; \alpha] \cap \Omega) \\ \lim_{\alpha \rightarrow -\infty} F(\alpha) &= 0 \\ \lim_{\alpha \rightarrow \infty} F(\alpha) &= 1 \end{aligned}$$

A function $f : \mathbb{R} \rightarrow \mathbb{R}^+$, such that

$$\int_{-\infty}^{\infty} f(t) dt = 1$$

is called **probability density function** (PDF). With any PDF we can associate a CDF

$$F(\alpha) = \int_{-\infty}^{\alpha} f(t) dt$$

and a probability measure $\mathbb{P}$ that associates the probability of the event $E \subset \mathbb{R}$ to occur with the area of all points underneath the function $f(t)$ whose t -values reside in E :

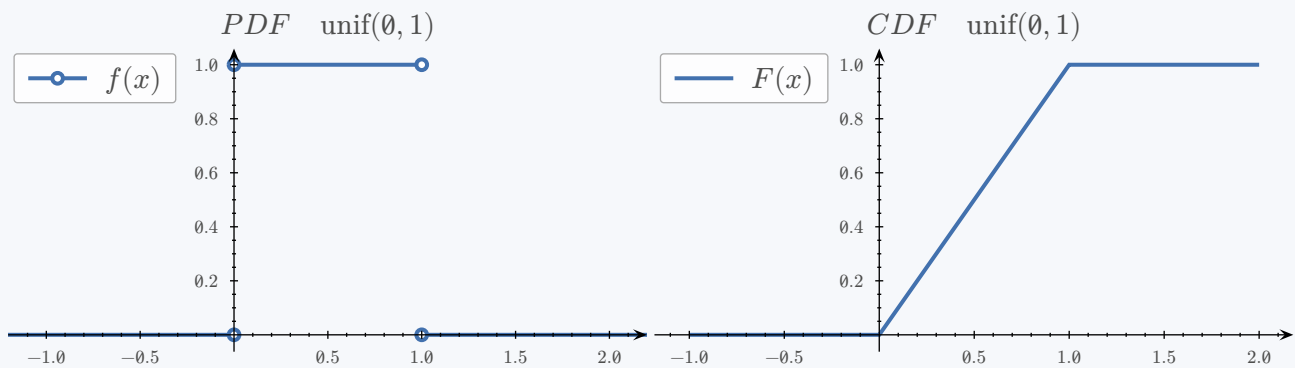
$$\mathbb{P}(E) = \int_{t \in E} f(t) dt$$

Example 4: Cumulative distribution function F and probability density function f for $\text{unif}(a, b)$

$$F : \begin{cases} \mathbb{R} \rightarrow [0; 1] \\ x \mapsto \mathbb{1}_{[a; b]}(x) \cdot \frac{x-a}{b-a} + \mathbb{1}_{(b; \infty)}(x) \end{cases}$$

$$F'(x) = f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{else} \end{cases} = \frac{\mathbb{1}_{(a; b)}(x)}{b-a} = \text{density for } X \sim \text{unif}(a, b)$$

$$\mathbb{P}(X \leq x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^x \frac{\mathbb{1}_{(a; b)}(t)}{b-a} dt \stackrel{a < x < b}{=} \int_a^x \frac{1}{b-a} dt = \frac{x-a}{b-a}$$



2.5. DISTRIBUTIONS

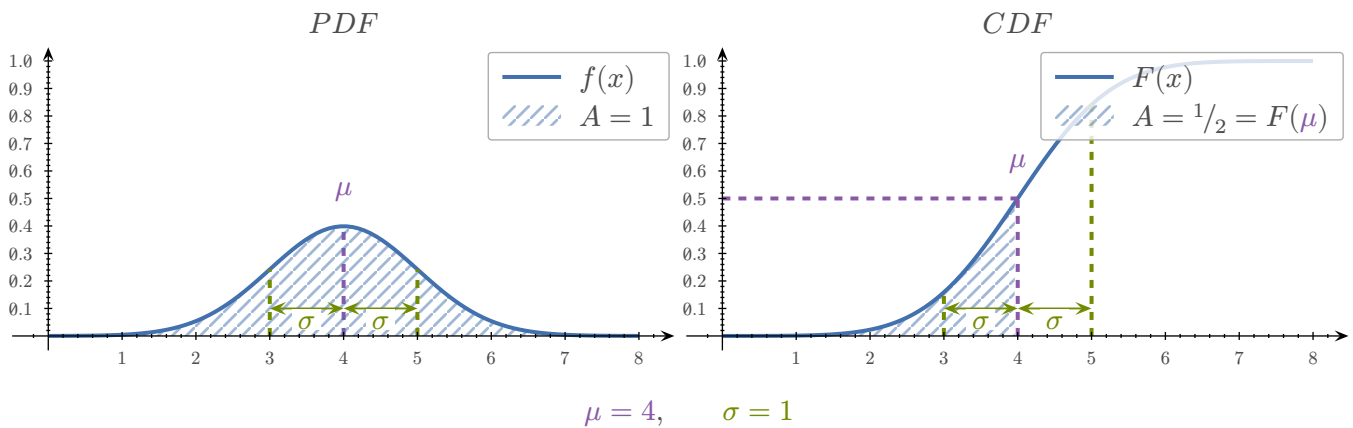
2.5.1. Univariate normal distribution

The probability density of the normal / Gaussian distribution is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

in which μ and σ are parameters, denoted as **mean value** (μ) and **standard deviation** (σ).

$$\int_{\mu-\sigma}^{\mu+\sigma} \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} dt \approx 0.68$$



The value of the CDF evaluated at x represents the area of the PDF from $-\infty$ to x , thus giving the probability of an experiment be in $(-\infty; x]$.

All normal distributions can be defined with the help of the so called **standard normal distribution**

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

for which $\mu = 0$ and $\sigma = 1$. The general normal distribution can then be expressed in terms of the standard normal distribution using the formula

$$f(x|\mu, \sigma) = \frac{1}{\sigma} \varphi\left(\frac{x - \mu}{\sigma}\right)$$

The **error function** is defined through

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

or its Taylor series

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(2k+1)} x^{2k+1}$$

and can be approximated through hyperbolic functions

$$\text{erf}(x) \approx \tanh\left(\frac{2}{\sqrt{\pi}}\left(x + \frac{11}{123}x^3\right)\right)$$

Given this function, one can show, that the CDF of the normal distribution is given by

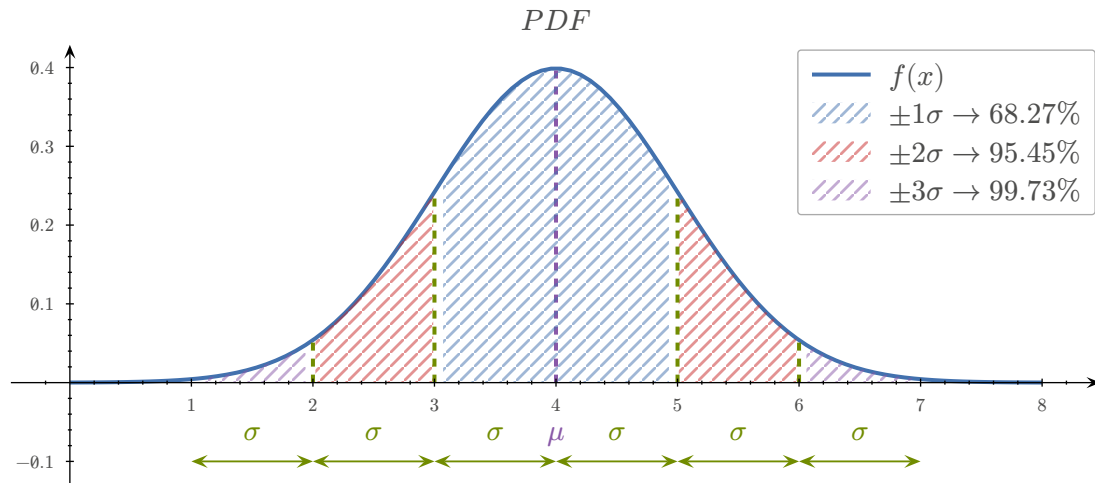
$$F(x|\mu, \sigma) = \frac{1}{2} \left(1 + \operatorname{erf} \left(\frac{x - \mu}{\sqrt{2\sigma^2}} \right) \right)$$

The **68–95–99.7 rule**, also known as the empirical rule, for a normal distribution is a shorthand used to remember the percentage of values that lie within an interval estimate in a normal distribution: approximately 68%, 95%, and 99.7% of the values lie within one, two, and three standard deviations of the mean, respectively.

$$\mathbb{P}(\mu - 1\sigma \leq X \leq \mu + 1\sigma) \approx 68.27\%$$

$$\mathbb{P}(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 95.45\%$$

$$\mathbb{P}(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 99.73\%$$



2.5.2. Bernoulli distribution

A random variable X is said to be a Bernoulli random variable with parameter p , shown as $X \sim \text{Bernoulli}(p)$, if its PMF is given by

$$P_X(x) = \begin{cases} p & \text{for } x = 1 \\ 1 - p & \text{for } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

where $0 < p < 1$

A Bernoulli random variable is associated with a certain event A . If event A occurs, then $X = 1$; otherwise $X = 0$. For this reason the Bernoulli random variable is also called the indicator random variable. The indicator random variable I_A for an event A is defined by

$$I_A = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

The indicator random variable for an event A has Bernoulli distribution with parameter $p = P(A)$, so we can write $I_A \sim \text{Bernoulli}(P(A))$.

TODO:

https://www.probabilitycourse.com/chapter3/3_1_5_special_discrete_distr.php

3. TREE DIAGRAMS AND SEQUENTIAL EVENTS

A **probabilistic model** that explains how data is generated (**generative model**). Generative Models connect domain knowledge with data.

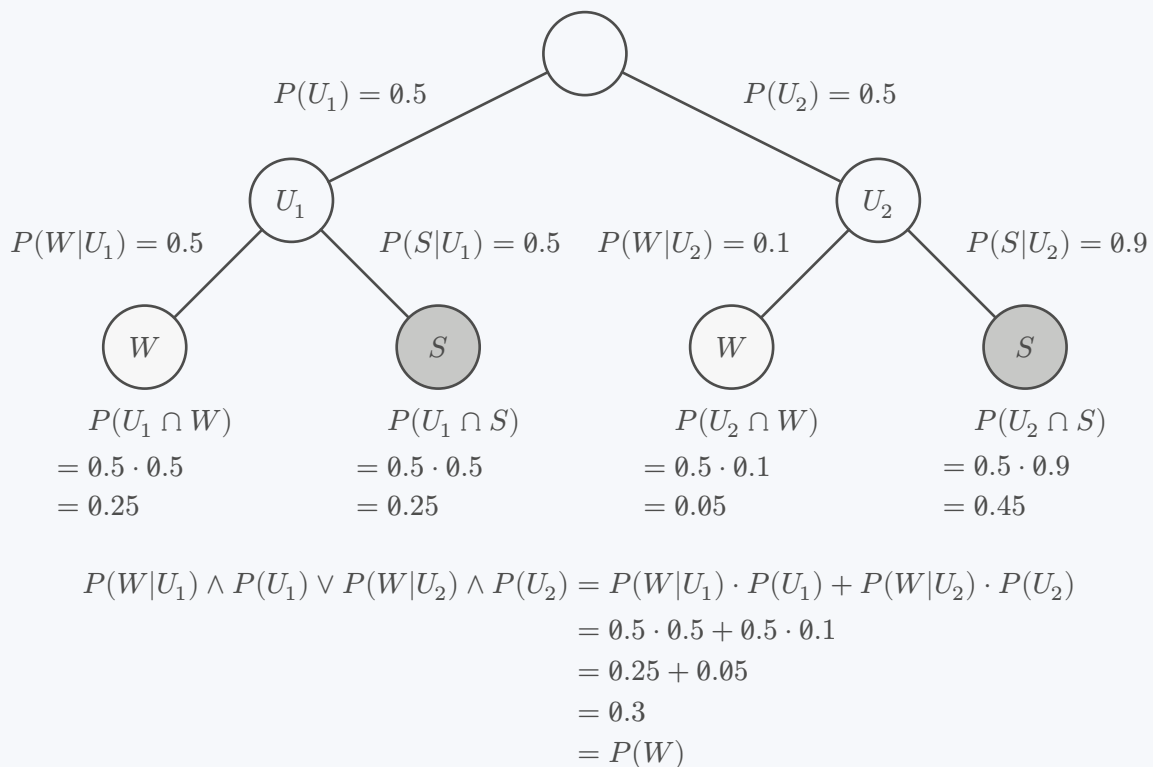
The 2-step model captures the causality of observable events.

- Step 1 lists all our **hypotheses** (mutually exclusive, exhaustive) that could have caused an outcome. (Example: U_1, U_2)
- Step 2 is an enumeration of all possible observable **consequences** (Example: W, S)

Example 5: Urns: Tree Diagram

There are two urns, U_1 and U_2 . Urn U_1 contains 5 silver and 5 white balls, and urn U_2 contains 9 silver and 1 white ball. The probability of drawing a ball from U_1 or U_2 is equal (50/50). What is the probability of drawing a white ball?

We can visualize the probabilities using a **tree diagram**:



The probability of drawing a white ball is therefore 30%, assuming that each urn is chosen with equal probability.

We can use Bayes' theorem to "go backwards" and start with the observation and deduce the cause.

$$P(\text{Hypothesis}|\text{Evidence}) = \frac{P(\text{Evidence}|\text{Hypothesis}) \cdot P(\text{Hypothesis})}{P(\text{Evidence})}$$

Bayesian inference is the statistical method that uses Bayes' theorem to update the probability of a hypothesis after seeing new data. It starts with a **prior probability** (belief before the evidence) and combines it with a likelihood to produce a **posterior probability** (updated belief).

Example 6: Urns: Bayesian inference

We have seen that the choice of urns yield following probabilities (prior):

$$P(U_1) = 0.5$$

$$P(U_2) = 0.5$$

If we observe the drawing of a white ball, we can deduce which urn was more likely to be chosen from (posterior):

$$P(U_1|W) = \frac{P(W|U_1) \cdot P(U_1)}{P(W)} = \frac{0.5 \cdot 0.5}{0.3} = 0.8\bar{3}$$

$$P(U_2|W) = \frac{P(W|U_2) \cdot P(U_2)}{P(W)} = \frac{0.5 \cdot 0.1}{0.3} = 0.1\bar{6}$$

Thus the **prior distribution** of equally distributed probabilities changes to the **posterior distribution** favoring U_1 when provided with the observed data.

Given multiple observed results we can use **Recursive Bayesian estimation** (also called **Bayesian filtering**), which is a general probabilistic approach for estimating an unknown PDF recursively.

$$p(x_0, \dots, x_k, z_1, \dots, z_k) = p(x_0) \prod_{i=1}^k p(z_i|x_i)p(x_i|x_{i-1})$$

Example 7: Urns: Recursive Bayesian estimation

Given the observed outcomes of white, silver, white ($\{W, S, W\}$), we can construct our posterior distribution recursively. We start with the prior $P(U_1) = 0.5$ and $P(U_2) = 0.5$.

1) After the first white ball (W):

$$P(U_1|W) \approx 0.833$$

$$P(U_2|W) \approx 0.167$$

2) Using these as the new priors, we observe a silver ball (S):

$$P(U_1|W, S) = \frac{P(S|U_1) \cdot 0.833}{P(S)} = \frac{0.5 \cdot 0.833}{0.5 \cdot 0.833 + 0.9 \cdot 0.167} \approx 0.73$$

$$P(U_2|W, S) = \frac{P(S|U_2) \cdot 0.167}{P(S)} = \frac{0.9 \cdot 0.167}{0.5 \cdot 0.833 + 0.9 \cdot 0.167} \approx 0.27$$

3) Using these as the new priors, we observe another white ball (W):

$$P(U_1|W, S, W) = \frac{0.5 \cdot 0.73}{0.5 \cdot 0.73 + 0.1 \cdot 0.27} \approx 0.93$$

$$P(U_2|W, S, W) = \frac{0.1 \cdot 0.27}{0.5 \cdot 0.73 + 0.1 \cdot 0.27} \approx 0.07$$

The evidence of two white balls strongly overwhelms the single silver ball, significantly increasing our confidence that the balls have been drawn from U_1 .